MINI PROJECT REPORT

Exploratory Data Analysis of the Early 2019-nCoV Outbreak (21–26 January 2020)

A visual EDA of the Johns Hopkins CSSE summary file
2019_nC0v_20200121_20200126 - SUMMARY.csv

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Registration Number	Ra2411056030023
Programme	Data Science and Machine Learning (DSML)
Project type	Module 1 Mini Project — Exploratory Data Analysis
Tools	Python, NumPy, Pandas, Matplotlib, Seaborn
Dataset window	21 January 2020 – 26 January 2020
Date of report	September 2026

*This report accompanies the Jupyter notebook **MiniProject_COVID19_nCoV_EDA.ipynb**. The analysis follows the same Load → Inspect → Clean → Explore → Visualise (12 charts) → Summarise workflow used in the course mini-project on India's COVID-19 data, applied here to the early 2019-nCoV summary file.*

1. Abstract

This mini project performs an exploratory data analysis of the earliest publicly collated 2019 novel coronavirus (2019-nCoV) situation reports. The source file, *2019_nC0v_20200121_20200126 - SUMMARY.csv*, is an aggregated version of the Johns Hopkins University CSSE dataset covering 21–26 January 2020. Using only NumPy, Pandas and Matplotlib/Seaborn, the analysis cleans mixed timestamps and missing counts, then tracks how confirmed cases grew from 332 to 2,794 in six days. Hubei province alone held 50.9% of confirmed cases by 26 January, Mainland China held 98.0%, and 17 locations outside Mainland China had already reported imported cases. Recovery rates were still below 2% because almost every patient was newly admitted. The findings are descriptive, not causal: they document the opening week of the outbreak and the data-quality problems that come with a brand-new pathogen.

2. Introduction and Objectives

On 31 December 2019 the Wuhan Municipal Health Commission reported a cluster of pneumonia cases of unknown cause. By the third week of January 2020 the pathogen had a name — 2019-nCoV — and Johns Hopkins CSSE had begun publishing province-level situation reports. Those first reports are noisy: blank numeric fields, trailing spaces in country names, and clocks written as “10pm”, “12am” or “23:00”. They are also historically important. They show the outbreak before it had a pandemic label, before India recorded its first case (30 January 2020), and long before vaccines existed.

The assignment is a Module 1 mini project in the same spirit as the course notebook *MiniProject_COVID19_India_EDA.ipynb*: load a real CSV, inspect it before touching it, clean it, aggregate with **groupby** / **pivot_table**, draw twelve charts, and write down what the charts actually say. The India notebook used MoHFW state-wise counts from the first wave (Jan–Aug 2020) plus OWID vaccination totals. This project uses the earlier JHU summary file specified for this submission, so the geographic grain is Chinese provinces plus the first international detections rather than Indian states.

Objectives

- Load and profile **2019_nC0v_20200121_20200126 - SUMMARY.csv**, including missingness and inconsistent labels.
- Clean dtypes, dates, country names and missing counts without leaking future information into the snapshot.
- Build daily global totals with groupby, and a country × date pivot of confirmed cases.
- Produce 12 visualisations covering cumulative trends, daily incidence, Hubei’s share, recovery/death rates, correlations, and international spread.
- Summarise what the first six days of reporting can — and cannot — tell us.

3. Dataset Description

The file is an aggregated stack of CSSE daily case updates from 21 through 26 January 2020 (ten snapshots, because some days published more than one bulletin). Each row is one location at one timestamp.

Column	Meaning	Notes after inspection
Province/State	Chinese province or sub-national area	Blank for national reports (Thailand, Japan, …)
Country	Country / region name	Trailing spaces on Singapore, Malaysia

Column	Meaning	Notes after inspection
Date last updated	Bulletin timestamp	Mixed text formats; parsed to datetime
Confirmed	Cumulative confirmed cases	Missing treated as 0
Suspected	Cases under investigation	Mostly missing after 25 Jan
Recovered	Cumulative recoveries	Very sparse — outbreak too new
Deaths	Cumulative deaths	Almost all in Hubei

Table 1. Raw columns in the 2019-nCoV summary file.

After stacking the bulletins the working table has **416 rows**, **19 countries/territories** and **40 named provinces**. A handful of Latin-American countries (Brazil, Mexico, Colombia) and the Philippines appear on the 23 January watchlist with zero confirmed cases; they are retained in the cleaned file but drop out of "latest confirmed" rankings because they still had no cases by 26 January.

4. Methodology

The workflow is the same six-step pipeline used in the course India EDA notebook, implemented in Python 3 with Pandas for tabular work and Matplotlib/Seaborn for charts.

4.1 Cleaning

Columns were renamed to snake_case. Country and province strings were stripped. Count fields were coerced with `pd.to_numeric(..., errors='coerce')` and missing values filled with 0 — a count that was not reported is treated as zero in these bulletins, not as "unknown but possibly large". Timestamps were parsed with `format='mixed'` so that "10pm", "12am" and 24-hour clocks land on the same datetime axis. An **active** column was derived as confirmed – deaths – recovered, floored at zero.

4.2 Aggregation

Because the CSV stacks every bulletin, summing every row would triple-count provinces that were updated twice on the same day. For each calendar date we keep only the last report per (province, country) pair, then sum. Daily new cases are the first difference of that cumulative series. A pivot table of confirmed cases by country and date shows when each country entered the file.

4.3 Visualisation design

Twelve charts were drawn to mirror the India mini-project layout: a national (here: global) cumulative line, a daily-new bar chart, a top-10 ranking, a recovery-rate ranking, a top-5 trend overlay, a share pie, a rate trend, a bubble scatter, a correlation heatmap, a boxplot of the distribution, a grouped comparison, and a "least / outside-core" bar chart. Colour is reserved for meaning (blue = confirmed, green = recovered, red = deaths, orange = outside China).

5. Data Cleaning Findings

Inspection of the raw file reproduced the same class of problems the India notebook had to fix (a text-typed death column, duplicate state spellings). Here the analogues were:

- **Missing counts.** Confirmed, suspected, recovered and deaths are blank on many early rows. Filling with 0 is conservative and matches how CSSE later published the series.

• **Inconsistent country strings.** “Singapore ” and “Malaysia ” with trailing spaces would have split those countries in a groupby.

• **Mixed clocks.** Pandas cannot infer a single format for “1/21/2020 10pm” and “1/26/20 23:00”; mixed parsing is required.

• **Watchlist rows.** Brazil, Mexico, Colombia and the Philippines appear with empty confirmed counts. They are not outbreaks; they are “no cases reported yet”.

• **Suspected column decays.** After 25 January the suspected field is largely abandoned as testing shifted the definition toward laboratory-confirmed cases.

6. Exploratory Analysis

6.1 Global snapshot

Metric	21 Jan 2020	26 Jan 2020
Confirmed cases	332	2,794
Deaths	6	80
Recovered	25	54
Active (derived)	301	2,660
Recovery rate	7.53%	1.93%
Death rate	1.81%	2.86%
Locations in bulletin	27	47
Countries / territories	7	15
Mainland China share of cases	—	98.0%
Hubei share of cases	—	50.9%

Table 2. Six-day change in the cleaned daily totals.

Confirmed cases grew by a factor of about 8.4 in six days. New cases per day were already approaching 1,000 by 26 January (+974). Recoveries barely moved, so the active-case curve tracks the confirmed curve. That is the signature of an outbreak that is still accelerating, not one that has peaked.

Date	Confirmed	New	Deaths	Recovered	CFR
21 Jan 2020	332	332	6	25	1.81%
22 Jan 2020	555	223	17	28	3.06%
23 Jan 2020	653	98	18	30	2.76%
24 Jan 2020	941	288	26	36	2.76%
25 Jan 2020	1,820	879	54	49	2.97%
26 Jan 2020	2,794	974	80	54	2.86%

Table 3. Daily global series after last-snapshot aggregation.

6.2 Chart 1 — Cumulative global trend

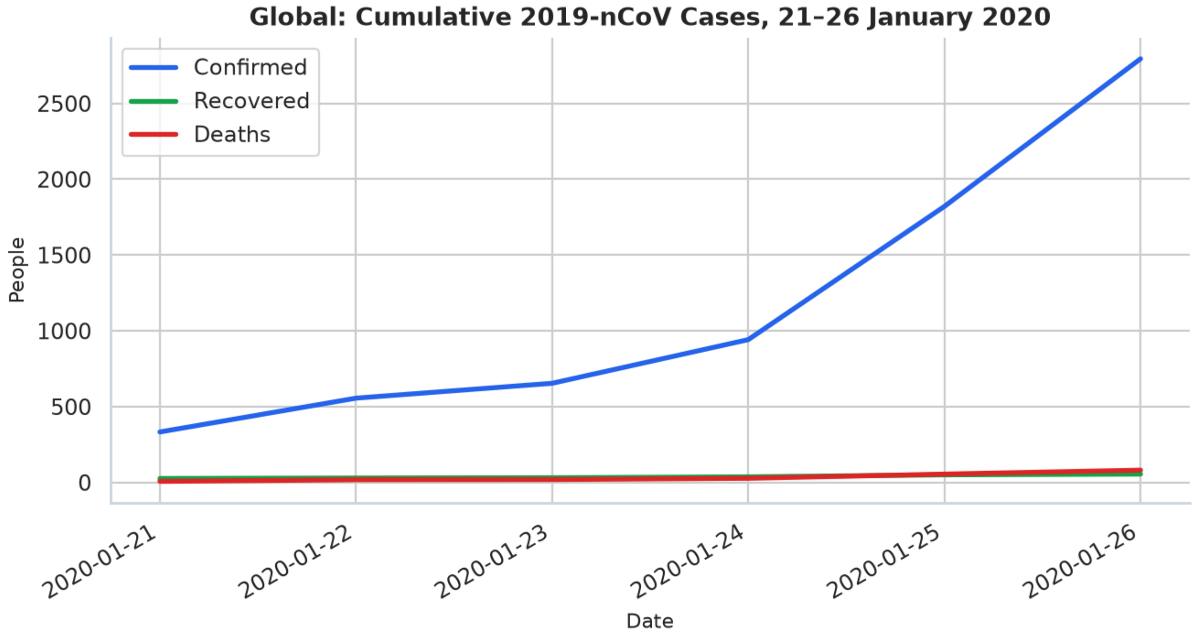

Figure 1. Cumulative confirmed, recovered and death counts, 21–26 January 2020.

The confirmed series is steep and convex. Recovered and death series are almost flat on this scale, not because outcomes were good, but because the time window is shorter than a typical hospital stay. This is the same cumulative-line chart used for India’s first wave in the course notebook; here the x-axis is days rather than months.

6.3 Chart 2 — Daily new confirmed cases

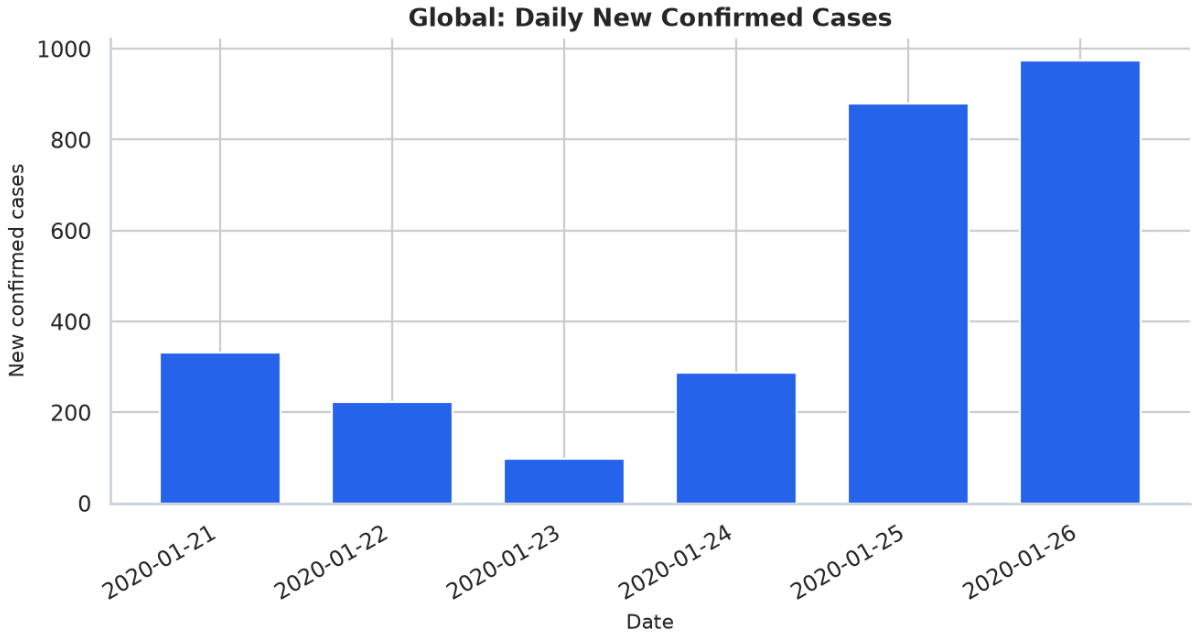

Figure 2. First difference of the global confirmed series.

Incidence is not smooth. 23 January is a quieter day (+98), then the next three bulletins jump to +288, +879 and +974. Part of that jump is genuine spread inside Hubei; part is improved case-finding after Wuhan’s lockdown was announced on 23 January. EDA cannot separate those two effects, and the report does not pretend to.

6.4 Chart 3 — Top 10 locations

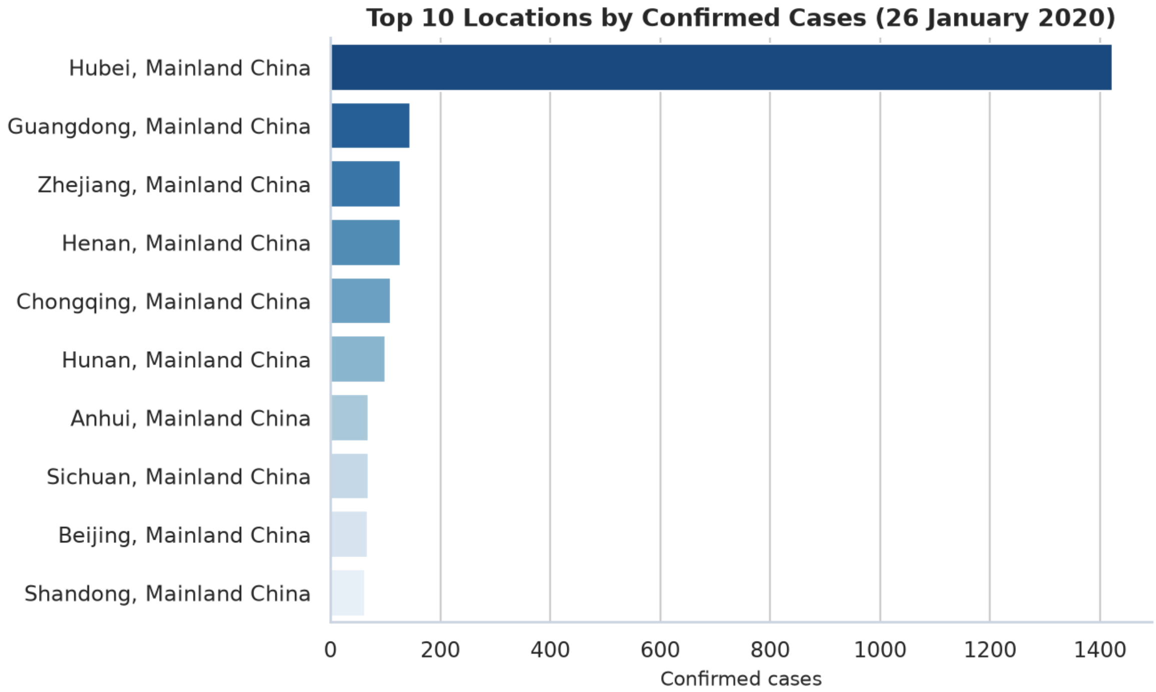

Figure 3. Top 10 locations by confirmed cases on 26 January 2020.

Location	Confirmed	Deaths	Recovered	CFR
Hubei, Mainland China	1,423	76	44	5.34%
Guangdong, Mainland China	146	0	2	0.00%
Zhejiang, Mainland China	128	0	1	0.00%
Henan, Mainland China	128	1	0	0.78%
Chongqing, Mainland China	110	0	0	0.00%
Hunan, Mainland China	100	0	0	0.00%
Anhui, Mainland China	70	0	0	0.00%
Sichuan, Mainland China	69	0	0	0.00%
Beijing, Mainland China	68	0	2	0.00%
Shandong, Mainland China	63	0	0	0.00%

Table 4. Ten worst-hit locations on the last bulletin of 26 January.

Hubei (1,423 confirmed, 76 deaths) sits in a different league from every other row. The next province, Guangdong, has roughly one-tenth as many cases and two recoveries. This is the provincial analogue of Maharashtra’s dominance in the India mini-project.

6.5 Chart 4 — Recovery rates of the worst-hit locations

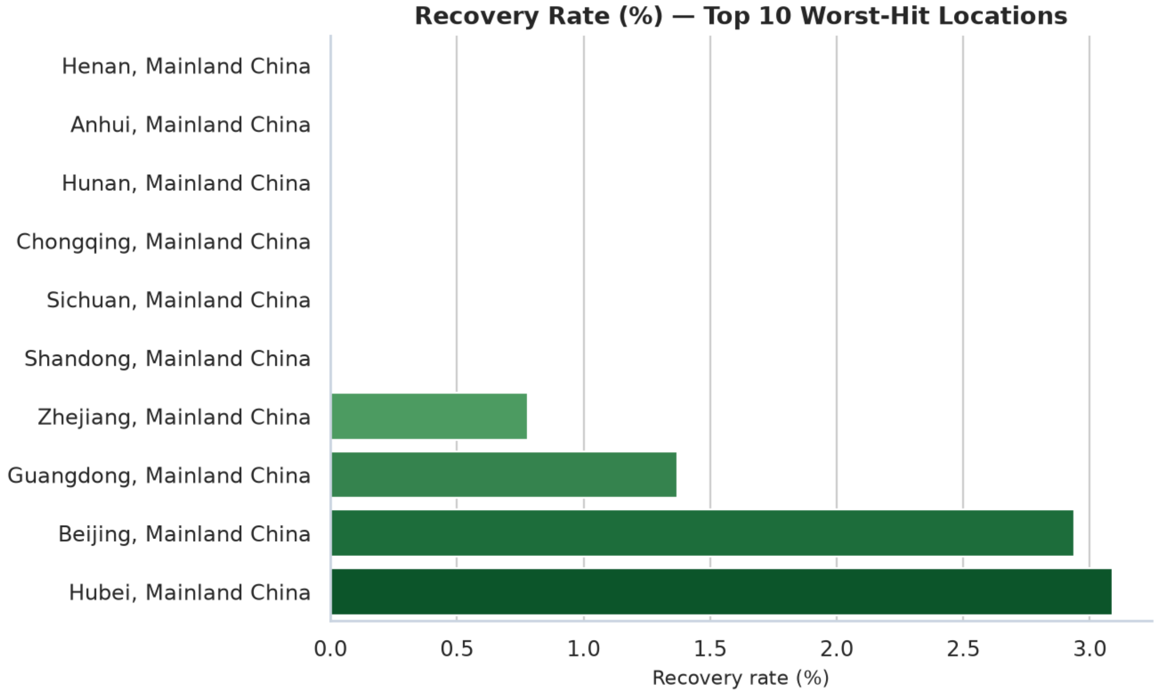

Figure 4. Recovery rate among the ten locations with the most confirmed cases.

Recovery rates are all in the low single digits. Beijing (2 recoveries / 68 cases) and Guangdong look "better" than Henan or Chongqing only because a couple of early patients had been discharged. With so few recoveries, ranking locations by recovery rate is unstable — a useful caution against over-interpreting ratios built on small numerators.

6.6 Chart 5 — Top-five trajectories

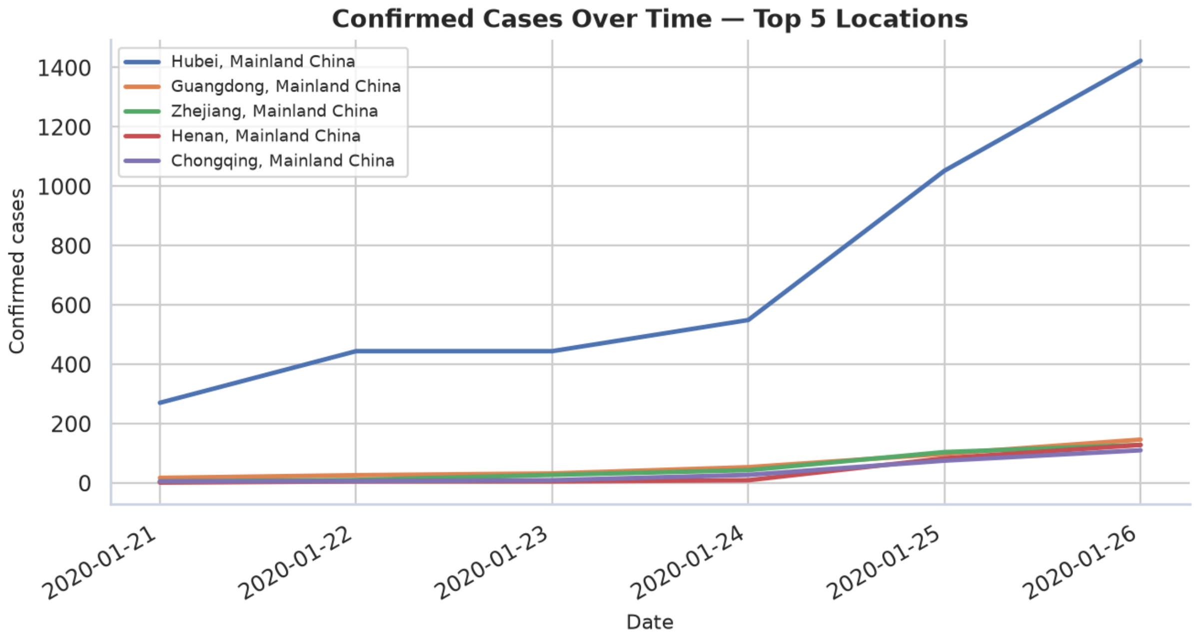

Figure 5. Confirmed-case trajectories for the five worst-hit locations.

Hubei’s curve bends upward after 24 January; the other four provinces rise linearly and stay an order of magnitude lower. If a single chart has to explain the first week of 2019-nCoV, it is this one: a provincial epicentre pulling away from the rest of the country.

6.7 Chart 6 — Share of global cases

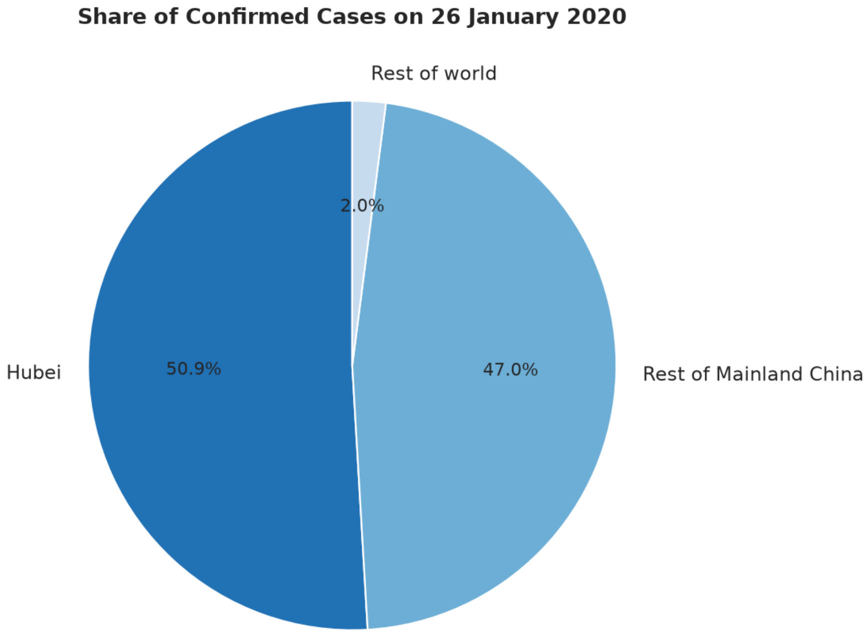

Figure 6. Hubei vs the rest of Mainland China vs the rest of the world.

On 26 January Hubei held 50.9% of confirmed cases, other Mainland Chinese provinces held 47.0%, and the rest of the world held only 2.0% (57 people). The pie is the counterpart of the “top 6 Indian states” share chart in the course notebook: the epidemic was geographically concentrated, not national-even.

6.8 Chart 7 — Recovery rate vs death rate over time

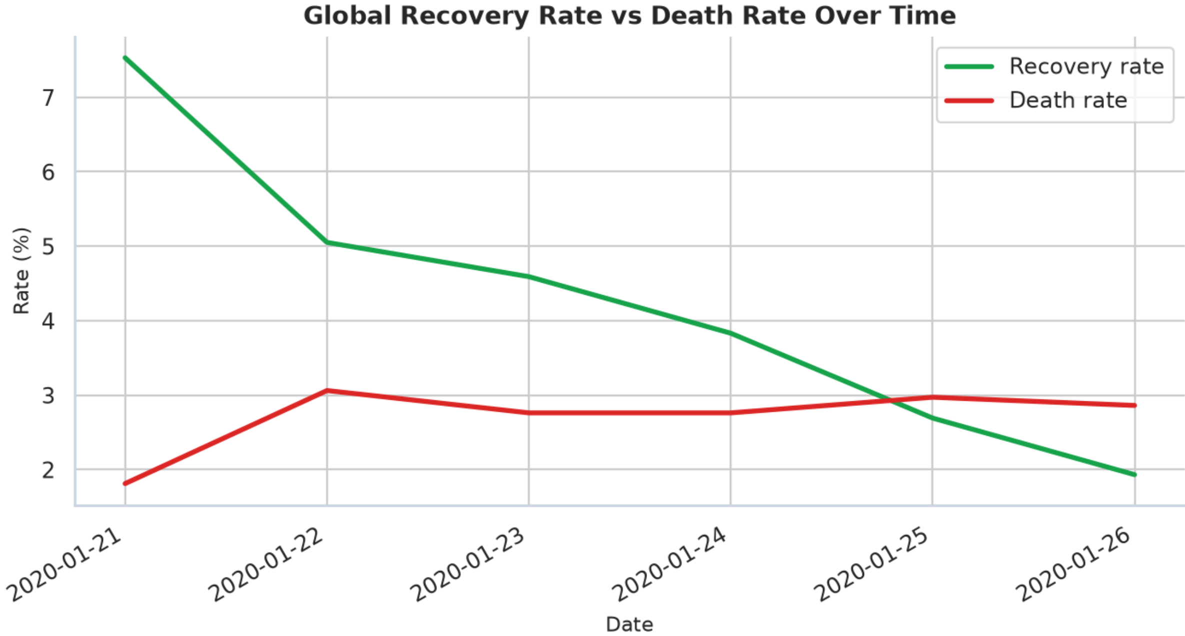

Figure 7. Crude recovery rate and crude death rate, global totals.

The crude death rate ends at 2.86% and the recovery rate at 1.93%. Both numbers will move as the denominator (confirmed) and the numerator (deaths, discharges) catch up with reality. They are reported here as the ratios the file actually contains, not as estimates of infection fatality.

6.9 Chart 8 — Deaths against confirmed cases

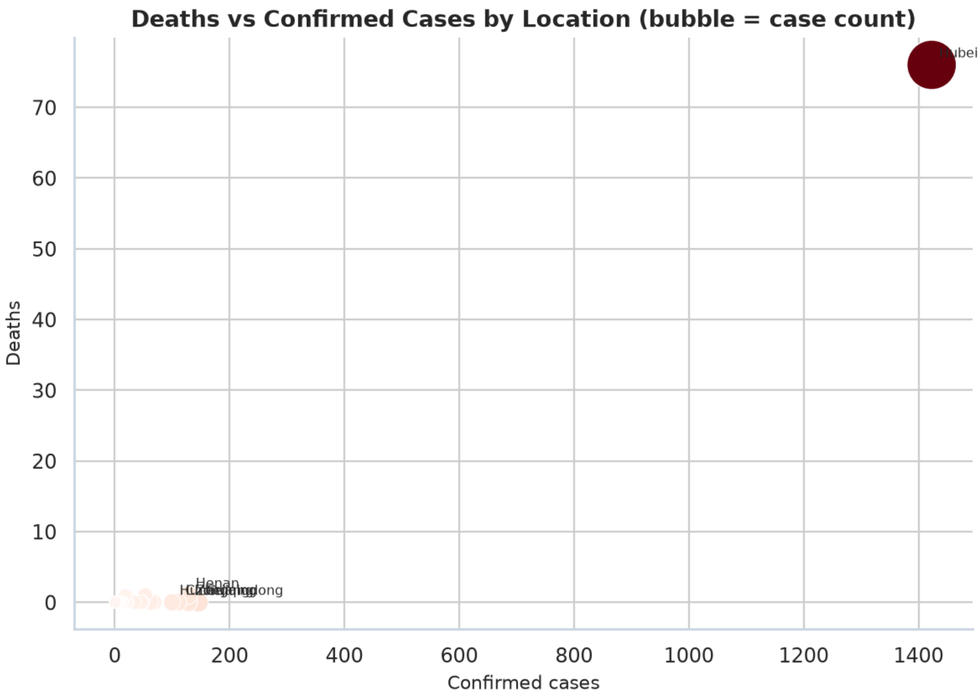

Figure 8. Location-level deaths vs confirmed cases. Bubble size is case count.

Hubei is the only point with a large death count. Everywhere else is still near the origin. Henan is the only other Mainland province with a recorded death in this window. The scatter is therefore a picture of concentration, not of a well-populated risk relationship.

6.10 Chart 9 — Correlation heatmap

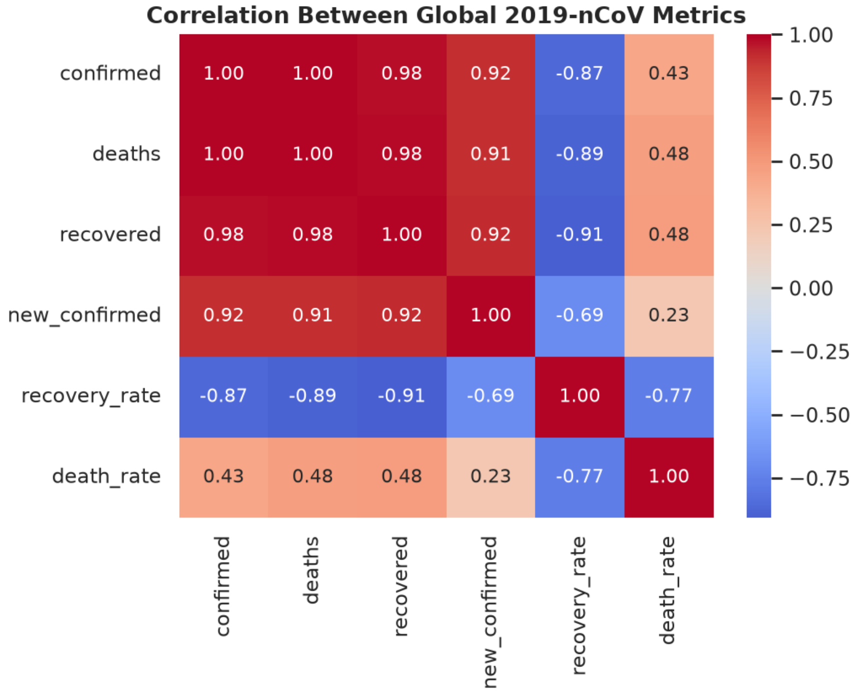

Figure 9. Pearson correlations among the six-day global series.

Confirmed, deaths, recovered and new_confirmed are strongly positively correlated because they are all rising together on a six-point series. Recovery rate is negatively related to the size metrics: as the confirmed denominator exploded, the recovered numerator did not keep up, so the ratio fell. With only six daily points the coefficients are descriptive of this window, not a general law.

6.11 Chart 10 — Distribution of location-level counts

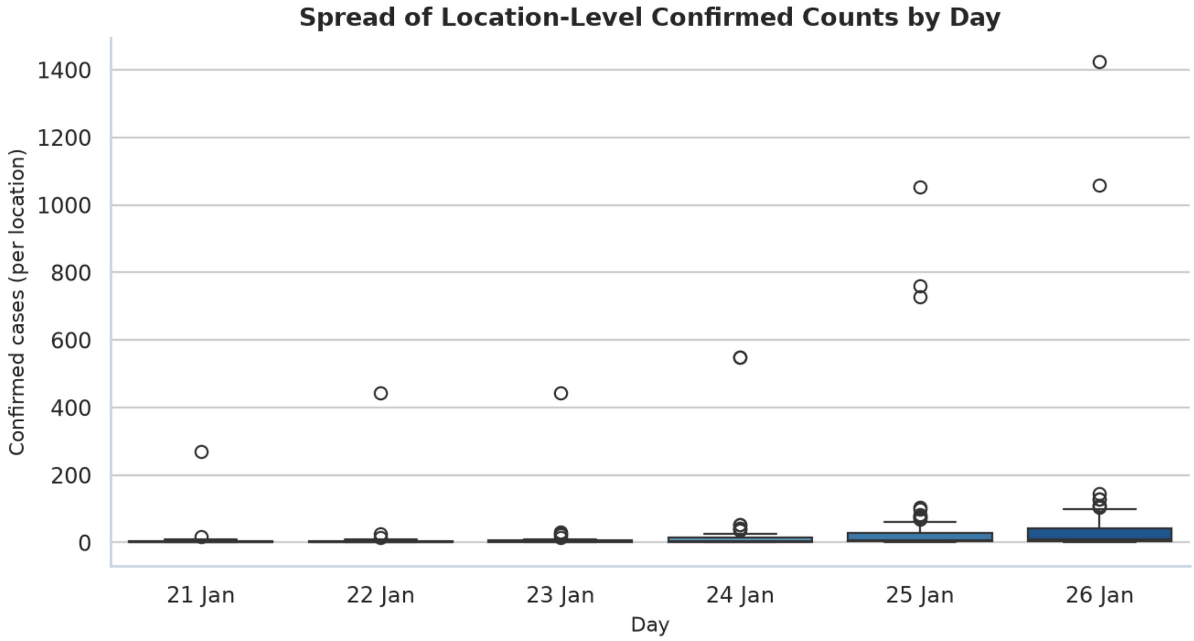

Figure 10. Boxplot of confirmed counts across locations, one box per day.

Each box is right-skewed with a high outlier — Hubei. The median location still has a modest count even on 26 January, which is another way of saying the mean is a Hubei story. Boxplots are used here the same way the India notebook used them for daily new cases by month: to show spread, not just the total.

6.12 Chart 11 — Mainland China versus the rest of the world

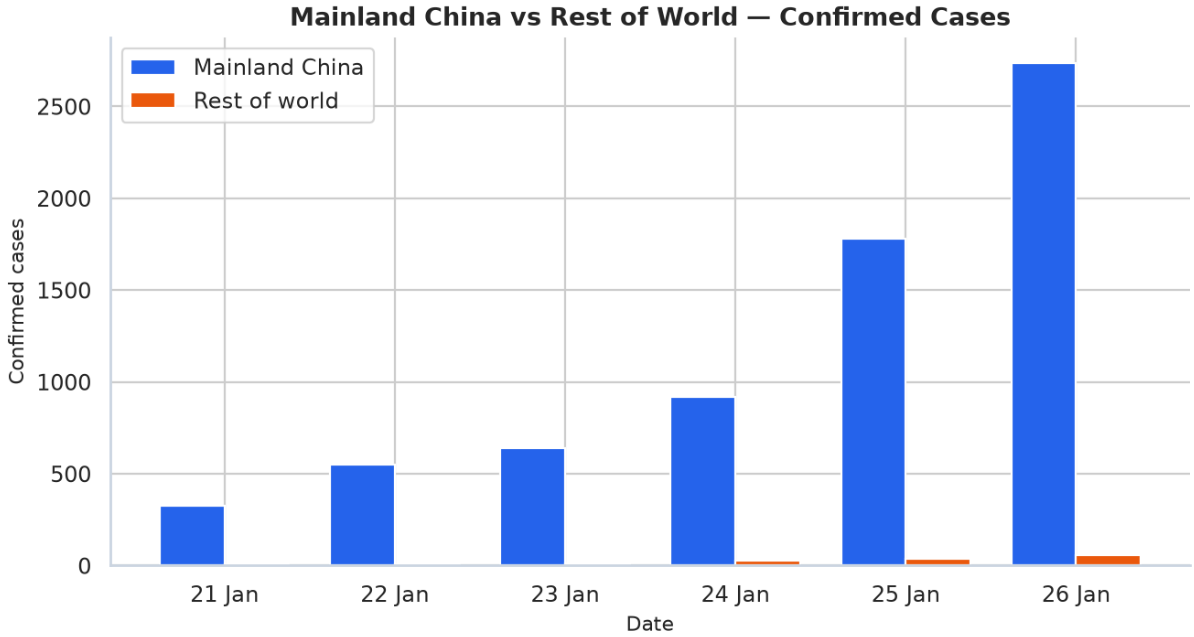

Figure 11. Grouped bars of confirmed cases inside vs outside Mainland China.

Outside Mainland China the series is almost invisible on the same axis. That is the point. On 26 January this was still a Chinese epidemic with a thin ring of imported detections, not a balanced global one. Replacing the vaccination chart from the India mini-project, this comparison is the "second chapter" the January 2020 file actually supports.

6.13 Chart 12 — International detections

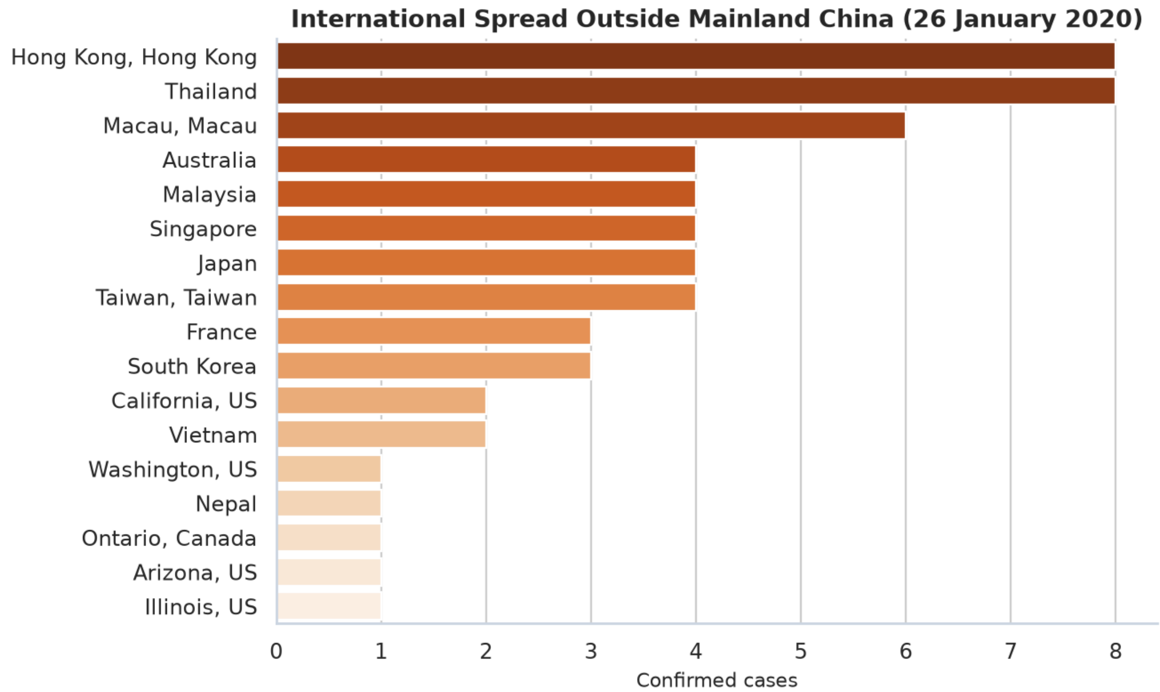

Figure 12. Confirmed cases outside Mainland China on 26 January 2020.

Location	Confirmed	Deaths	Recovered
Hong Kong, Hong Kong	8	0	0
Thailand	8	0	2
Macau, Macau	6	0	0
Australia	4	0	0
Malaysia	4	0	0
Singapore	4	0	0
Japan	4	0	1
Taiwan, Taiwan	4	0	0
France	3	0	0
South Korea	3	0	0
California, US	2	0	0
Vietnam	2	0	0
Washington, US	1	0	0
Nepal	1	0	0
Ontario, Canada	1	0	0
Arizona, US	1	0	0
Illinois, US	1	0	0

Table 5. Every location outside Mainland China with a 26 January bulletin.

The largest non-Mainland counts are Hong Kong and Thailand (8 each). The United States has 5, split across Washington, Illinois, California and Arizona in the underlying rows. Nepal’s single case is the first detection on the subcontinent in this file; India itself is not yet present. No deaths are recorded outside Mainland China in this window.

7. Key Findings

- **Eight-fold growth in six days.** Confirmed cases moved from 332 (21 Jan) to 2,794 (26 Jan), with daily new cases reaching 974 on the last day.
- **Hubei was the outbreak.** 1,423 of 2,794 confirmed cases (50.9%) and 76 of 80 deaths sat in one province.
- **Mainland China held 98.0% of cases.** The rest of the world had 57 confirmed people, all of them living, all of them in single-digit national totals.
- **Recovery had not started in earnest.** Only 54 recoveries against 2,794 confirmed cases (1.93%). Low recovery here is a lag, not a clinical conclusion.
- **The crude death rate was 2.86%.** It is dominated by Hubei and will be biased by under-testing of mild cases.
- **International spread was already real, and still tiny.** 14 locations outside Mainland China were on the board; none had reached double-digit deaths, and India had not yet reported a case.
- **The raw CSV needed the same cleaning discipline as the India dataset:** missing counts, dirty strings, mixed dates, and watchlist rows that are not cases.

8. Comparison with the Course India EDA

The course notebook *MiniProject_COVID19_India_EDA.ipynb* analysed MoHFW state-wise counts from 30 January to 6 August 2020 (India's first wave) and an OWID vaccination series from 2021–24. This submission keeps that pedagogical skeleton and swaps in the specified early-outbreak file.

Item	Course India EDA	This 2019-nCoV EDA
File	complete.csv + India OWID vax	2019_nC0v_20200121_20200126 - SUMMARY.csv
Window	30 Jan – 6 Aug 2020	21–26 Jan 2020
Grain	Indian state / UT	Chinese province + country
Cleaning	Death as text; Telangana spellings	Missing counts; trailing spaces; mixed clocks
Headline concentration	Maharashtra ~¼ of India	Hubei ~51% of the world file
Recovery	~68% by Aug 2020	1.93% (too early to discharge)
Vaccination chart	2.2 bn doses by 2024	Not applicable — replaced by China vs world
India in the file	Entire subject	Not yet present (first case 30 Jan)

Table 6. How this mini project maps onto the course EDA template.

9. Limitations

These bulletins are not a complete epidemic curve. Testing capacity in mid-January 2020 was low, case definitions were still moving, and suspected counts disappear from the schema within the week. Crude rates ignore reporting lag. Locations with zero confirmed cases remain in the raw file and must not be plotted as outbreaks. Nothing in this analysis is a forecast, a treatment effect, or a ranking of health-system quality.

10. Conclusion

The first six published days of 2019-nCoV data already contain the shape of what followed: exponential growth, extreme geographic concentration in Hubei, a death count that had started to move, a recovery count that had not, and a thin but real ring of international importations. Cleaning the summary file is not a preface to the analysis — it is part of the analysis. Once the dates parse and the country names stop splitting, twelve charts are enough to tell that story with the Module 1 toolkit.

Submitted by **Advik Singh**, registration number **Ra2411056030023**. The executable companion to this report is **MiniProject_COVID19_nCoV_EDA.ipynb**.

11. References

[1] Johns Hopkins University Center for Systems Science and Engineering (CSSE). 2019 Novel Coronavirus COVID-19 (2019-nCoV) Data Repository. Archived daily case updates, 21–26 January 2020.

[2] Dong, E., Du, H. & Gardner, L. (2020). An interactive web-based dashboard to track COVID-19 in real time. *The Lancet Infectious Diseases*.

[3] Dey, S. K., Rahman, M. M., Siddiqi, U. R. & Howlader, A. (2020). Analyzing the epidemiological outbreak of COVID-19: A visual exploratory data analysis approach. *Journal of Medical Virology*. Uses 2019_nC0v_20200121_20200126-SUMMARY.csv as a primary source.

[4] Course material: MiniProject_COVID19_India_EDA.ipynb — Exploratory Data Analysis on India's COVID-19 Data (Module 1 template: NumPy, Pandas, Matplotlib/Seaborn).

[5] Our World in Data; Ministry of Health and Family Welfare, Government of India — referenced only as the data sources of the course India notebook, not used in this run.

Appendix A. Reproducibility

Place `2019_nC0v_20200121_20200126 - SUMMARY.csv` next to the notebook and run all cells top to bottom. Dependencies are listed in `requirements.txt` (pandas, numpy, matplotlib, seaborn). The notebook writes `cleaned_ncov_summary.csv` as a reusable output. This PDF was generated from the same cleaned tables and the twelve figures in the `figures/` folder.

Student	**Declaration**
Advik Singh Ra2411056030023	This mini project is my own work, prepared with the specified dataset and the course EDA template.